Osptek Display

OLED SPECIFICATION

Model No:

OED192-128160W001-C31

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REVISION RECORD

REV.	REVISION DESCRIPTION	REV. DATE	REMARK
X01	■ INITIAL RELEASE	2018. 05. 14	

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1. SCOPE

The purpose of this specification is to define the general provisions and quality requirements that apply to the supply of display cells manufactured by Osptek. This document, together with the Module Assembly Drawing, is the highest-level specification for this product. It describes the product, identifies supporting documents and contains specifications, which are either not addressed, or are exceptions to the supporting documents.

2. WARRANTY

Osptek warrants that the products delivered pursuant to this specification (or order) will conform to the agreed specifications for twelve (12) months from the shipping date ("Warranty Period"). Osptek is obligated to repair or replace the products which are found to be defective or inconsistent with the specifications during the Warranty Period without charge, on condition that the products are stored in the original packages at 25℃±5℃, 55%±10%RH or used as the conditions specified in the specifications.
Nevertheless, Osptek is not obligated to repair or replace the products without charge if the defects or inconsistency are caused by the force majeure or the reckless behaviors of the customer.
After the Warranty Period, all repairs or replacements of the products are subject to charge.

3. FEATURES

- Small molecular organic light emitting diode.
- Color : White
- Panel resolution : 128x160
- Driver IC : SH1108
- Excellent quick response time.
- Extremely thin thickness for best mechanism design : 1.427 mm
- High contrast : 2,000:1
- Wide viewing angle : 160°
- Strong environmental resistance.
- 8-bit 6800-series parallel interface, 8-bit 8080-series parallel interface, and 3-wire & 4-wire serial peripheral interface, I^2C bus interface.
- Wide range of operating temperature : -40 to 70℃
- Anti-glare polarizer.

4. MECHANICAL DATA

NO	ITEM	SPECIFICATION	UNIT
1	Dot Matrix	128 x 160	dot
2	Dot Size	0.206 (W) x 0.226 (H)	mm^2
3	Dot Pitch	0.226 (W) x 0.246 (H)	mm^2
4	Aperture Rate	84	%
5	Active Area	28.908 (W) x 39.34 (H)	mm^2
6	Panel Size	32.9 (W) x 46.55 (H)	mm^2
7*	Panel Thickness	1.22 ± 0.15	mm
8	Module Size	32.9 (W) x 118.4 (H) x 1.427 (T)	mm^3
9	Diagonal A/A size	1.92	inch
10	Module Weight	TBD	gram

* Panel thickness includes substrate glass, cover glass and UV glue thickness.

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5. MAXIMUM RATINGS

ITEM	MIN	MAX	UNIT	Condition	Remark
Supply Voltage (V_{DD})	-0.3	3.6	V	Ta = 25℃	IC maximum rating
Supply Voltage (V_{PP})	8	17	V	Ta = 25℃	IC maximum rating
Operating Temp.	-40	70	℃	-	-
Storage Temp	-40	85	℃	-	Note (2)

Note:

(1) Maximum ratings are those values beyond which damages to the OLED module may occur. The OLED functional operation should be restricted to the limits in the section 6. Electrical Characteristics tables.

(2) The defined temperature ranges do not include the polarizer. The maximum withstood temperature of the polarizer should be 80℃.

6. ELECTRICAL CHARACTERISTICS

6.1 D.C ELECTRICAL CHARACTERISTICS

SYMBOL	PARAMETERS	TEST CONDITION	MIN	TYP	MAX	UNIT
V_{PP}	Analog power supply (for OLED panel)	Ta 25℃	13	13.5	14	V
V_{DD}	Digital power supply	Ta 25℃	1.65	2.85	3.5	V
V_{IH}	High logic input level		0.8* V_{DD}	-	V_{DD}	V
V_{IL}	Low logic input level		V_{SS}	-	0.2* V_{DD}	V
V_{OH}	High logic output level	I_{OH} = -0.5mA	0.8* V_{DD}	-	V_{DD}	V
V_{OL}	Low logic output level	I_{OL} = 0.5mA	V_{SS}	-	0.2* V_{DD}	V

Note: The V_{PP} input must keep in a stable value; ripple and noise are not allowed.

6.2 ELECTRO-OPTICAL CHARACTERISTICS

PANEL ELECTRICAL SPECIFICATIONS

PARAMETER	MIN	TYP.	MAX	UNITS	COMMENTS
Normal mode current (IPP)	-	66	68	mA	All pixels on (1)
	-	16	17	mA	20% pixels on (1)
Standby mode current (IPP)	-	5	6	mA	Standby mode 10% pixels on (2)
Normal mode power consumption	-	891	918	mW	All pixels on (1)
	-	216	229.5	mW	20% pixels on (1)
Standby mode power consumption	-	67.5	81	mW	Standby mode 10% pixels on (2)
IDD sleep mode current	-	-	5	uA	Sleep mode Current (3)
IPP sleep mode current	-	-	5	uA	Sleep mode Current (3)
Normal mode Luminance	115	145		cd/m^2	Display Average
Standby Luminance		65		cd/m^2	
CIEx (White)	0.32	0.36	0.40		CIE1931
CIEy (White)	0.34	0.38	0.42		CIE1931
Dark Room Contrast	2000:1				
Viewing Angle	160			degree	
Response Time		10		μs	

(1) Normal mode condition :

- Driving Voltage : 13.5V
- Contrast setting : 0xd0
- Frame rate : 105Hz
- Duty setting : 1/128

(2) Standby mode condition :

- Driving Voltage : 13.5V
- Contrast setting : 0x00
- Frame rate : 105Hz
- Duty setting : 1/128

(3) Sleep mode condition :

When send 0xae command OLED display off and memory data will be maintained.

(4) Wake up condition :

When send 0xaf command OLED will be turned on.

7. LIFETIME SPECIFICATION

ITEM	MIN	UNIT	Condition	Remark
Life Time	10,000	Hrs	145 cd/m^2, 50% checkerboard	Note (1)
Life Time	13,000	Hrs	115 cd/m^2, 50% checkerboard	Note (2)

Note:

(A) Under VPP = 13.5V, Ta = 25℃, 50% RH.

(B) Life time is defined the amount of time when the luminance has decayed to less than 50% of the initial measured luminance.

(1) Setting of 145 cd/m^2 :

- Contrast setting : 0xd0
- Frame rate : 105Hz
- Duty setting : 1/128

(2) Setting of 115 cd/m^2 :

- Contrast setting : 0x80
- Frame rate : 105Hz
- Duty setting : 1/128

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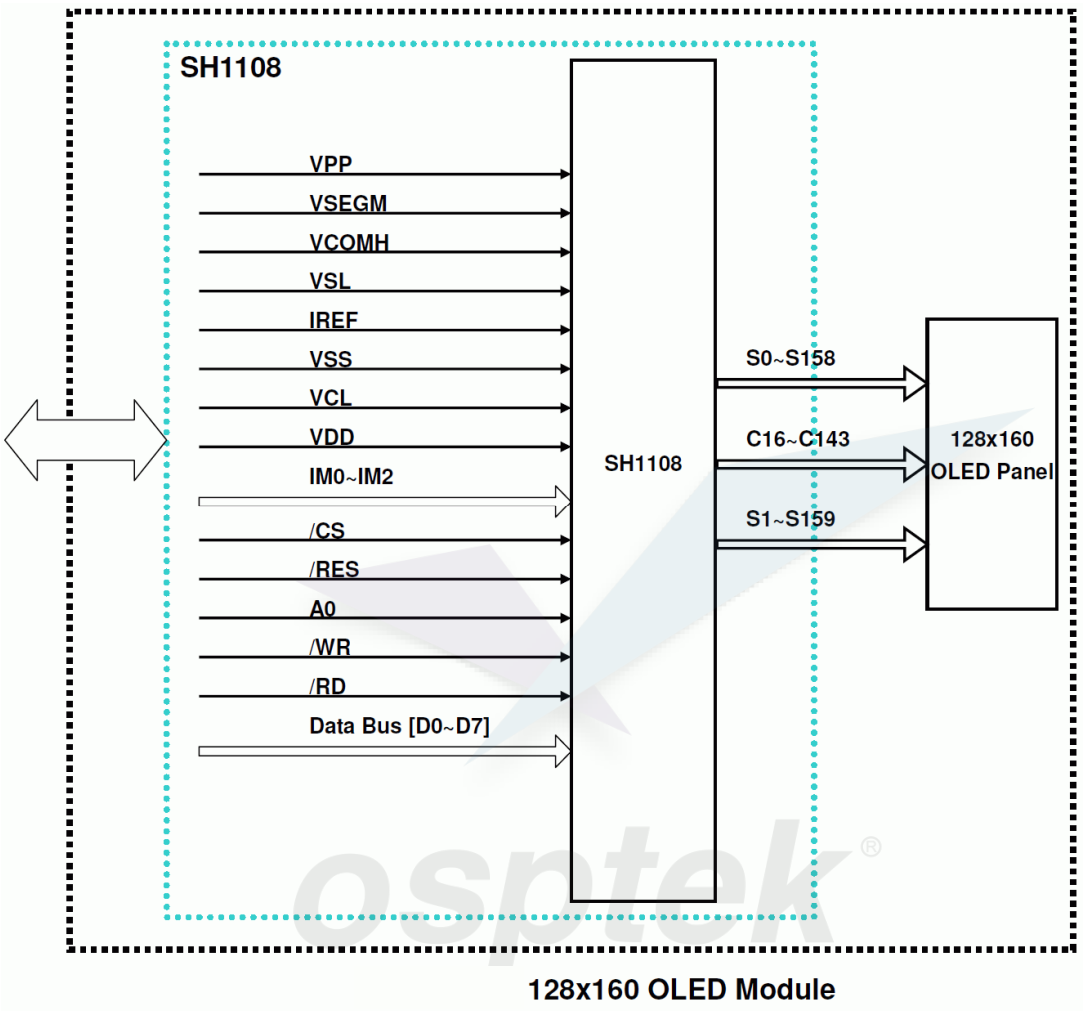
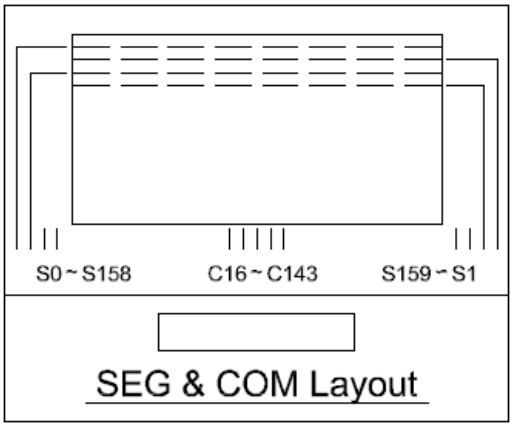
8. INTERFACE

8.1 FUNCTION BLOCK DIAGRAM



8.2 PANEL LAYOUT DIAGRAM



8.3 PIN ASSIGNMENTS

Pin No.	Pin Name	Description	Setting at each interface		
			8080 parallel	4-Wire SPI	IIC
1	NC	No connection.			
2	VPP	Power supply for panel driving voltage.			
3	VSEGM	This is voltage output high level for segment pre-charge. A capacitor should be connected between this pin and VSS.			
4	VCOMH	This is voltage output high level for common signals. A capacitor should be connected between this pin and VSS.			
5	VSL	This is a segment voltage reference pin. A capacitor should be connected between this pin and VSS.			
6	NC	No connection.			
7	IREF	This is a segment current reference pin. A resistor should be connected between this pin and VSS.			
8	VPP	Power supply for panel driving voltage.			
9	NC	No connection.			
10	VSS	Ground pin.			
11	VCL	This is a common voltage reference pin. This pin should be connected to VSS externally.			
12	VDD	Power supply for logic and input/output.			
13	IM0	This is MPU interface mode select pin.	Low	Low	Low
14	IM1	This is MPU interface mode select pin.	High	Low	High
15	VDD	Power supply for logic and input/output.			
16	IM2	This is MPU interface mode select pin.	High	Low	Low
17	/CS	This pin is the chip select input. When CS = “L”, then the chip select becomes active,and data/command I/O is enabled.	CS#	CS#	Low
18	/RES	This is a reset signal input pin. When RES is set to “L”, the settings are initialized.			
19	A0	This is the Data/Command control pin that determines whether the data bits are data or a command. A0 = “H”: the inputs at D0 to D7 are treated as display data. A0 = “L”: the inputs at D0 to D7 are transferred to the command registers. In I^2C interface, this pin serves as SA0 to distinguish the different address of OLED driver.	A0	A0	SA0
20	/WR	This is a MPU interface input pin. When connected to an 8080 MPU, this is active LOW. This pin connects to the 8080 MPU WR signal. The signals on the data bus are latched at the rising edge of	/WR	Low	Low

		the WR signal. When connected to a 6800 Series MPU: This is the read/write control signal input terminal. When R/W = “H”: Read. When R/W = “L”: Write.			
21	/RD	This is a MPU interface input pin. When connected to an 8080 series MPU, it is active LOW. This pin is connected to the RD signal of the 8080 series MPU, and the data bus is in an output status when this signal is “L”. When connected to a 6800 series MPU, this is active HIGH. This is used as an enable clock input of the 6800 series MPU.	/RD	Low	Low
22	D0	This is an 8-bit bi-directional data bus that connects to an 8-bit or 16-bit standard MPU data bus. When the serial interface is selected, then D0 serves as the serial clock input pad (SCL) and D1 serves as the serial data input pad (SI). At this time, D2 to D7 are set to high impedance. When the I^2C interface is selected, then D0 serves as the serial clock input pad (SCL) and D1 serves as the serial data input pad (SDA). At this time, D2 to D7 are set to high impedance.	D0	SCL	SCL
23	D1		D1	SI	SDA
24	D2		D2	Low	Low
25	D3		D3	Low	Low
26	D4		D4	Low	Low
27	D5		D5	Low	Low
28	D6		D6	Low	Low
29	D7		D7	Low	Low
30	NC	No connection.			
31	VPP	Power supply for panel driving voltage.			

Note

(1) Low is connected to VSS

(2) High is connected to VDD

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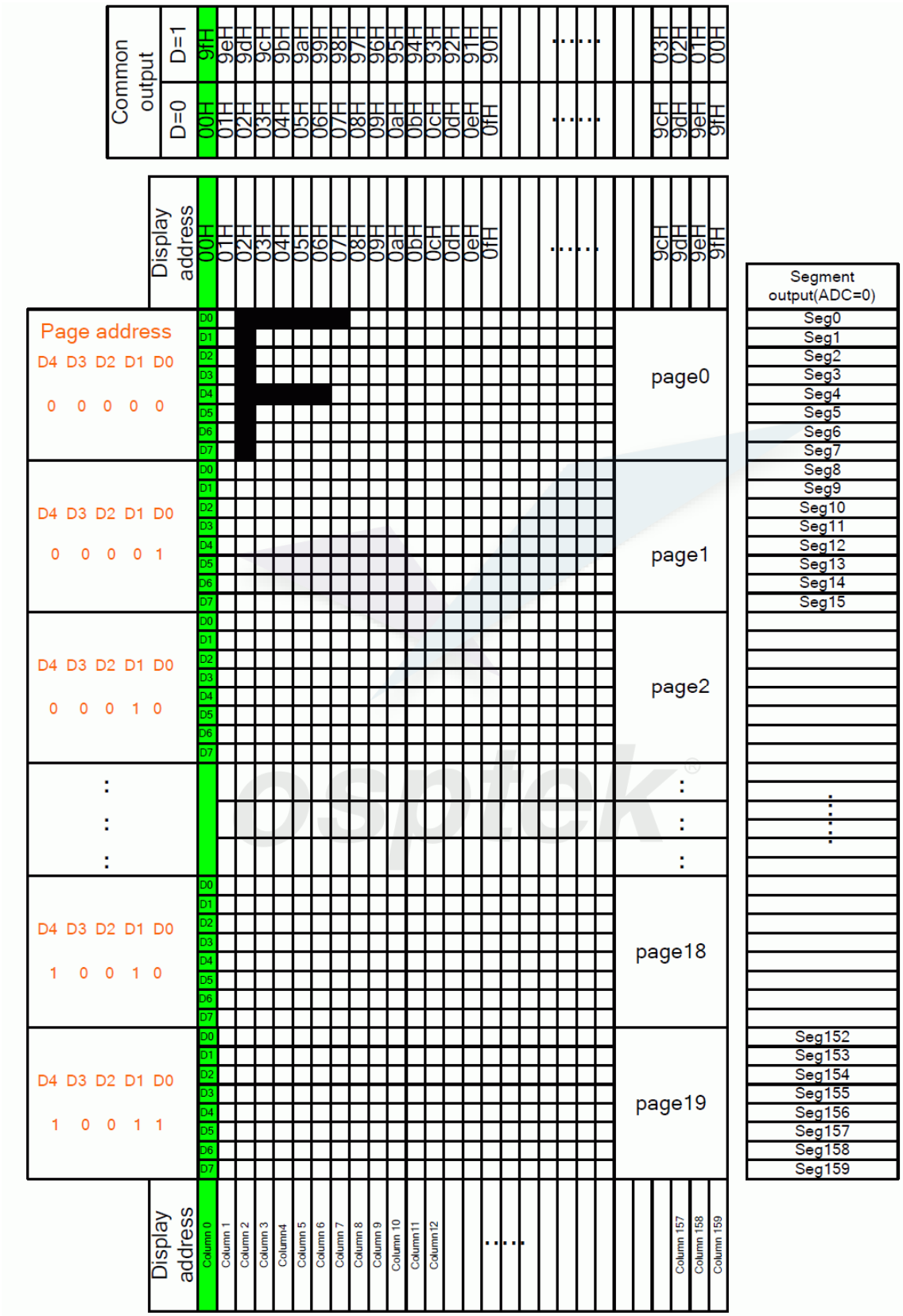
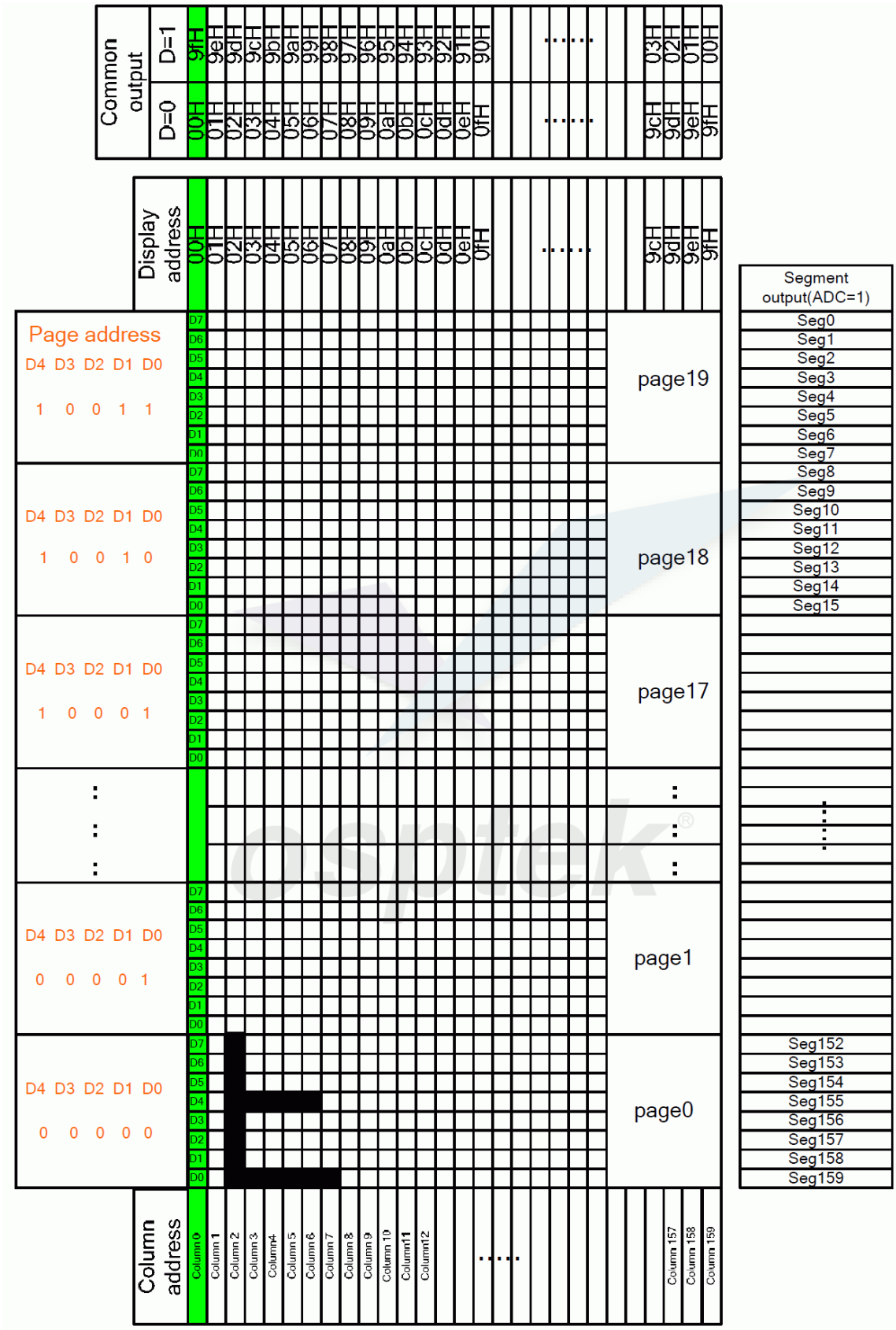
8.4 GRAPHIC DISPLAY DATA RAM ADDRESS MAP



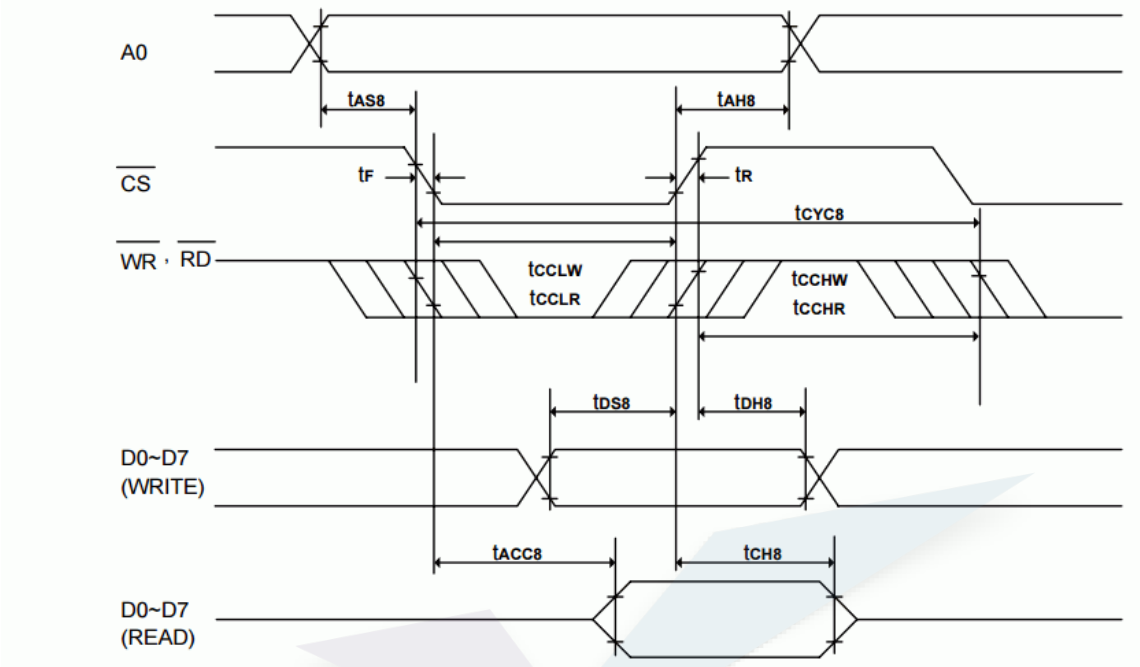
RAM map example: ADC=0(POR)



RAM map example: ADC=1(Seg remap)

8.5 INTERFACE TIMING CHART

System buses Read/Write characteristics 1 (For the 8080 Series Interface MPU)



(V_{DD} = 1.65V – 2.4V, T_A = +25°C)

Symbol	Parameter	Min.	Typ.	Max.	Unit	Condition
t_{CYC8}	System cycle time	300	-	-	ns	
t_{AS8}	Address setup time	0	-	-	ns	
t_{AH8}	Address hold time	0	-	-	ns	
t_{DS8}	Data setup time	40	-	-	ns	
t_{DH8}	Data hold time	30	-	-	ns	
t_{CH8}	Output disable time	10	-	70	ns	C_L = 100pF
t_{ACC8}	$\overline{RD}$ access time	-	-	280	ns	C_L = 100pF
t_{CCLW}	Control L pulse width (WR)	100	-	-	ns	
t_{CCLR}	Control L pulse width (RD)	120	-	-	ns	
t_{CCHW}	Control H pulse width (WR)	100	-	-	ns	
t_{CCHR}	Control H pulse width (RD)	100	-	-	ns	
t_R	Rise time	-	-	15	ns	
t_F	Fall time	-	-	15	ns	

(V_{DD} = 2.4V – 3.5V, T_A = +25°C)

Symbol	Parameter	Min.	Typ.	Max.	Unit	Condition
t_{CYC8}	System cycle time	300	-	-	ns	
t_{AS8}	Address setup time	0	-	-	ns	
t_{AH8}	Address hold time	0	-	-	ns	
t_{DS8}	Data setup time	40	-	-	ns	
t_{DH8}	Data hold time	15	-	-	ns	
t_{CH8}	Output disable time	10	-	70	ns	C_L = 100pF
t_{ACC8}	$\overline{RD}$ access time	-	-	140	ns	C_L = 100pF
t_{CCLW}	Control L pulse width (WR)	100	-	-	ns	
t_{CCLR}	Control L pulse width (RD)	120	-	-	ns	
t_{CCHW}	Control H pulse width (WR)	100	-	-	ns	
t_{CCHR}	Control H pulse width (RD)	100	-	-	ns	
t_R	Rise time	-	-	15	ns	
t_F	Fall time	-	-	15	ns	

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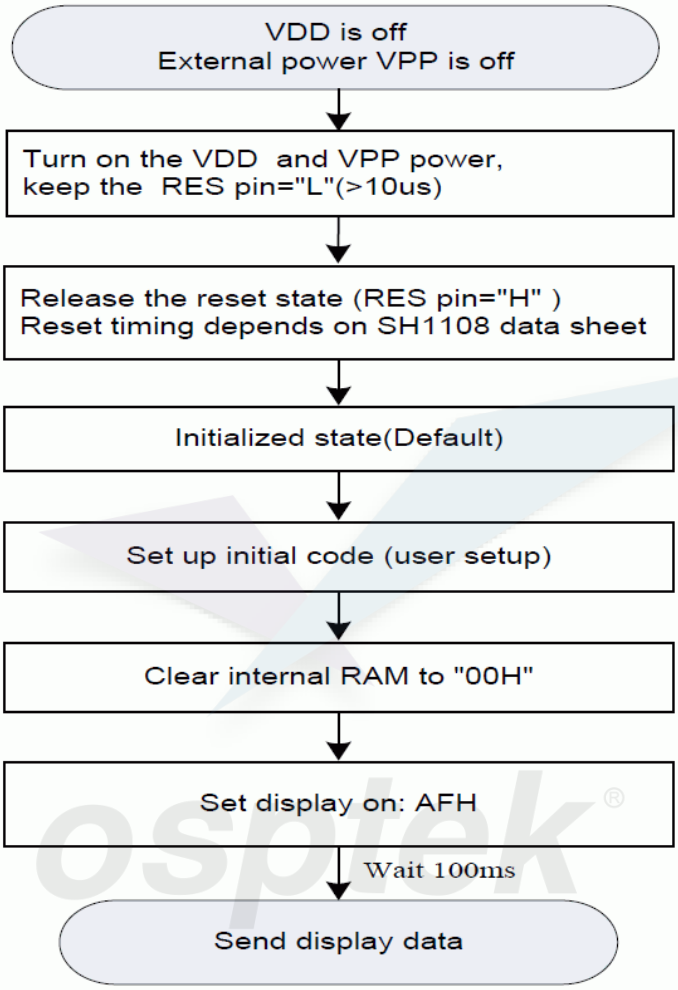
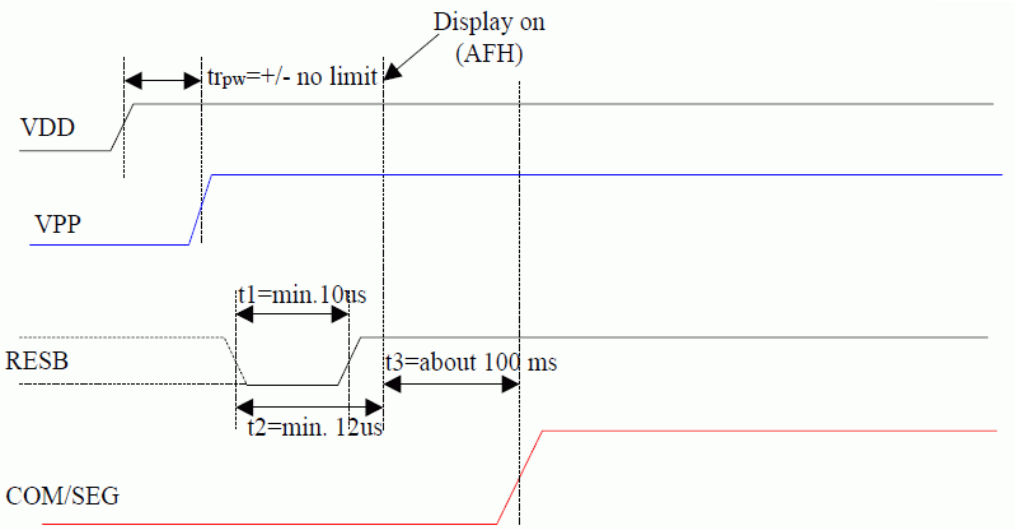
9. POWER ON / OFF SEQUENCE & APPLICATION CIRCUIT

9.1 POWER ON / OFF SEQUENCE

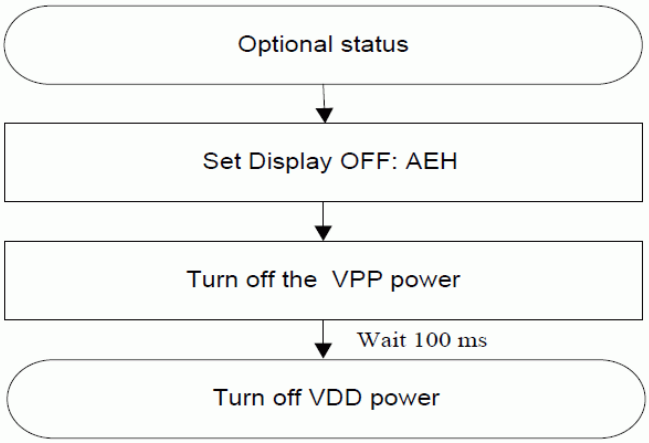
Power on sequence:

External power is being used immediately after turning on the power:





Power off sequence:



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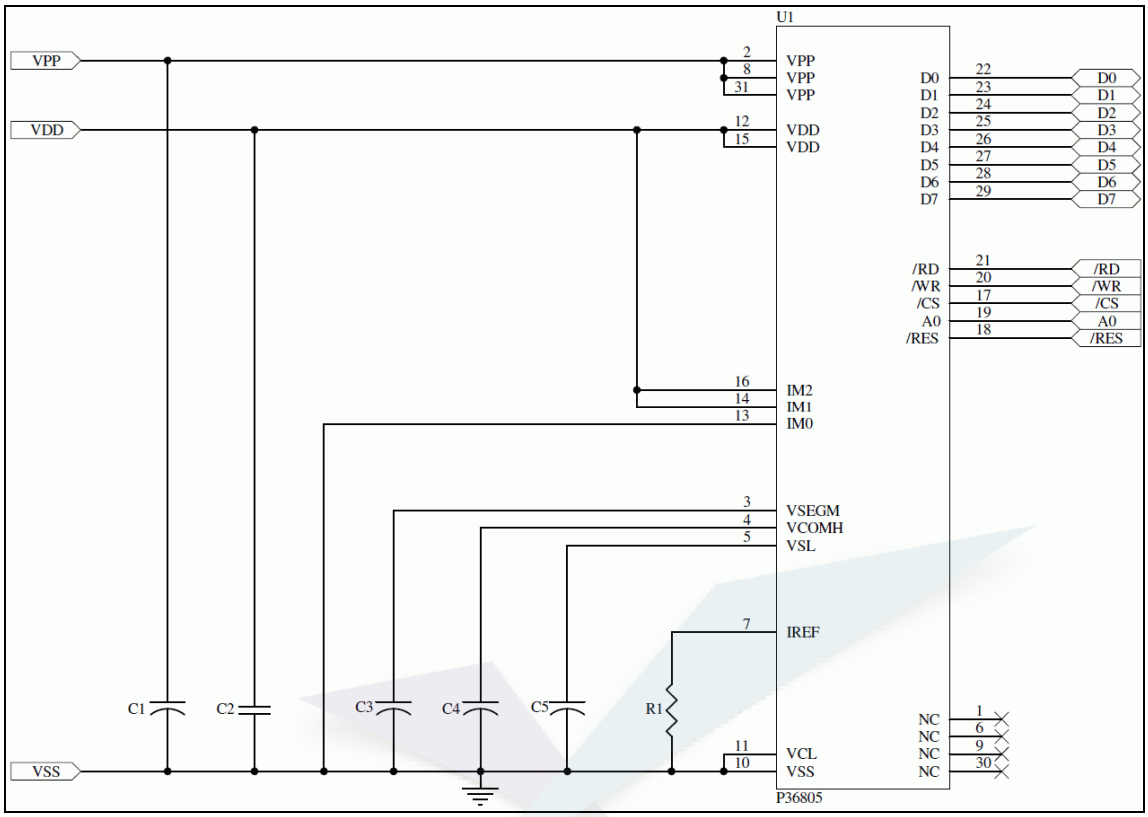
9.2 APPLICATION CIRCUIT



Recommend components：

C1 ,C3 ,C4, C5：4.7uF/25V VISHAY (572D475X0025A2T) or (Tantalum type)

C2：1uF/6.3V(0603)

R1: 750K ohm 1%(0603)

This circuit is for 8080 8bits interface

9.3 COMMAND TABLE

Refer to IC Spec.: SH1108G

10. RELIABILITY TEST CONDITIONS

No.	Items	Specification	Quantity
1	High temp. (Non-operation)	85℃, 240hrs	5
2	High temp. (Operation)	70℃, 120hrs	5
3	Low temp. (Operation)	-40℃, 120hrs	5
4	High temp. / High humidity (Operation)	65℃, 90%RH, 120hrs	5
5	Thermal shock (Non-operation)	-40℃ ~85℃ (-40℃ /30min; transit /3min; 85℃ /30min; transit /3min) 1cycle: 66min, 100 cycles	5
6	Vibration	Frequency : 5~50HZ, 0.5G Scan rate : 1 oct/min Time : 2 hrs/axis Test axis : X, Y, Z	1 Carton
7	Drop	Height: 120cm Sequence : 1 angle、3 edges and 6 faces Cycles: 1	1 Carton
8	ESD (Non-operation)	Air discharge model, ±8kV, 10 times	5

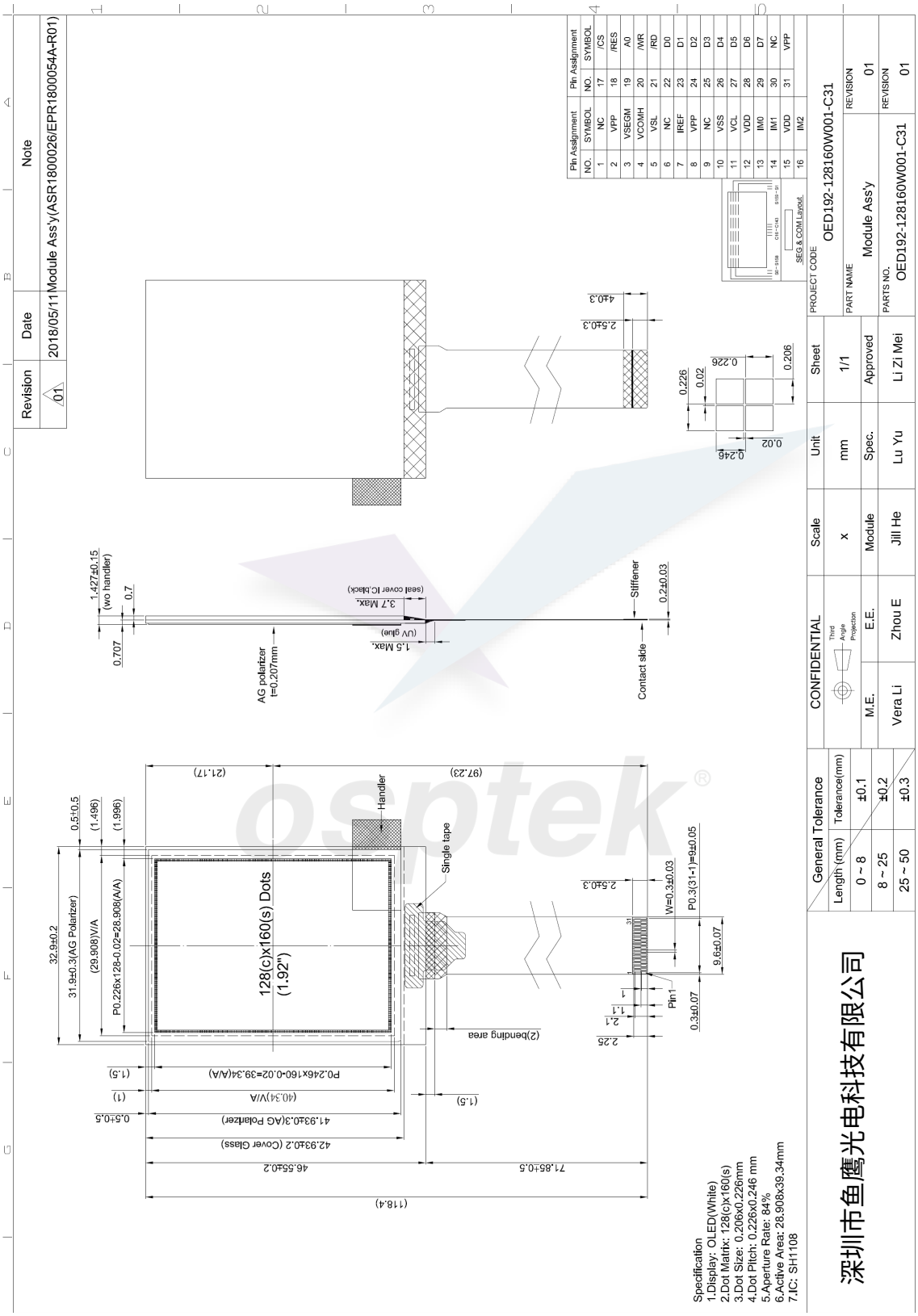
Test and measurement conditions

1. All measurements shall not be started until the specimens attain to temperature stability.
2. The degradation of Polarizer are ignored for item 1, 4 & 5.

Evaluation criteria

1. The function test is OK.
2. No observable defects.
3. Luminance: > 50% of initial value.
4. Current consumption: within ± 50% of initial value.

11. EXTERNAL DIMENSION



12. PACKING SPECIFICATION

TBD

13. OUTGOING INSPECTION PROVISION

1. 抽樣方法 / SAMPLING METHOD

(1) MIL-STD-1916 / 驗證水準 level III / 正常檢驗 / 單次樣品檢驗
MIL-STD-1916 / inspection level III / normal inspection / single sample inspection

(2) 主要缺陷 Level III；次要缺陷 Level II
Major Level III ; Minor Level II

MIL-STD-1916樣本代字對照表							
批量	驗證水準（VL）						
	VII	VI	V	IV	III	II	I
2～170	A	A	A	A	A	A	A
171～288	A	A	A	A	A	A	B
289～544	A	A	A	A	A	B	C
545～960	A	A	A	A	B	C	D
961～1632	A	A	A	B	C	D	E
1633～3072	A	A	B	C	D	E	E
3073～5440	A	B	C	D	E	E	E
5441～9216	B	C	D	E	E	E	E
9217～17408	C	D	E	E	E	E	E
17409～30720	D	E	E	E	E	E	E
≧30721	E	E	E	E	E	E	E

2. 檢驗條件 / INSPECTION CONDITION

檢查和測量在下列條件下進行的，除非另有規定。
The inspection and meaurement are performed under the following conditions, unless otherwise specified.

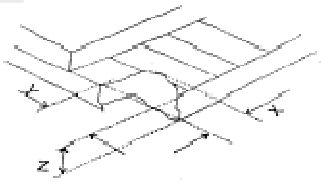
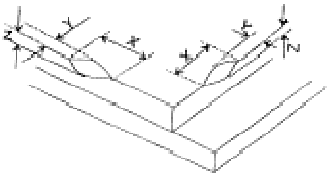
溫度 / Temperature: 25±5℃
濕度 / Humidity: 50±10%R.H.
壓力 / Pressure: 860~1060hPa (mbar)
檢驗員拿的面板和眼睛之間的距離 / Distance between the panel and eyes of the inspector≧30cm

3. 品質檢驗規格 / SPECIFICATION FOR QUALITY CHECK

3.1 缺陷分類 / DEFECT CLASSIFICATION

嚴重度 Severity	檢驗項目 Inspection Item	缺陷 Defect	備註 Remark
主要缺陷 Major Defect	1. 面板 Panel	(1) 無顯示 Non-displaying	
		(2) 線缺陷 Line defects	
		(3) 故障 Malfunction	
		(4) 玻璃破損 Glass cracked	
	2. 軟板 Film	(1) 軟板尺寸超規 Film dimension out of specification	不能組裝 Can not be assembled
	3. 尺寸 Dimension	(1) 外形尺寸超規 Outline dimension out of specification	
次要缺陷 Minor Defect	1. 面板 Panel	(1) 玻璃刮傷 Glass scratch	外觀缺陷 Appearance defect
		(2) 玻璃切割異常 Glass cutting NG	
		(3) 玻璃崩邊、崩角 Glass chip	
	2. 偏光板 Polarizer	(1) 偏光板刮傷 Polarizer scratch	
		(2) 表面汙漬 Stains on surface	
		(3) 偏光板氣泡 Polarizer bubbles	
	3. 顯示 Displaying	(1) 暗點、亮點、髒污 Dim spot、Bright spot、dust	
	4. 軟板 Film	(1) 損傷 Damage	
		(2) 異物 Foreign material	

3.2 出貨規格 / OUTGOING SPECIFICATION

<table>
<tr><th>項目
Item</th><th>描述
Description</th><th>標準
Criterion</th><th>允收
水準
AQL</th></tr>
<tr><td rowspan="3">I. 面板
Panel</td><td>1. 玻璃刮傷
Glass scratch</td><td>
<table>
<tr><th>寬 / Width
(mm)
W</th><th>長 / Length
(mm)
L</th><th>容許個數
number of pieces permitted</th></tr>
<tr><td>W≦0.03</td><td>忽略
Ignore</td><td>忽略
Ignore</td></tr>
<tr><td>0.03＜ W≦0.05</td><td>L≦1</td><td>1</td></tr>
<tr><td>0.05＜ W</td><td>------</td><td>無
None</td></tr>
<tr><td>顯示區外
beyond A.A.</td><td>------</td><td>忽略
Ignore</td></tr>
</table>
</td><td>次要
Minor</td></tr>
<tr><td>2. 玻璃破損
Glass crack</td><td>(1) 裂紋 / Crack
擴展裂紋是不能接受的。
Propagation crack is not acceptable.
 </td><td>主要
Major</td></tr>
<tr><td>3. 玻璃崩邊、崩角
Glass chip</td><td>(1) 崩角 / Chip on corner
 
(2) 崩邊 / Chip on edge
  </td><td>次要
Minor</td></tr>
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<table>
<tr><th>項目
Item</th><th>描述
Description</th><th>標準
Criterion</th><th>允收
水準
AQL</th></tr>
<tr><td rowspan="2">I. 面板
Panel</td><td>3. 玻璃崩邊、崩角
Glass chip</td><td>
<table>
<tr><th>崩角
Chip on corner</th><th>Size
(mm)</th><th>崩邊
Chip on edge</th><th>Size
(mm)</th></tr>
<tr><td>X</td><td>≦1.5</td><td>X</td><td>≦3.0</td></tr>
<tr><td>Y</td><td>≦2.0</td><td>Y</td><td>≦1.0</td></tr>
<tr><td>Z</td><td>≦t</td><td>Z</td><td>≦t</td></tr>
</table>
備註 / Note:
1. t = 玻璃厚度
t = glass thickness
2. 崩邊或崩角延伸到 ITO 導線是不能接受的。
Chip on the corner extending into the ITO contact is not acceptable.
</td><td>次要
Minor</td></tr>
<tr><td>4. 尺寸
Dimension</td><td>請參閱圖紙的規範。
Refer to the drawing of the spec</td><td>主要
Major</td></tr>
<tr><td rowspan="3">II. 偏光板
Polarizer</td><td>1. 刮傷
Scratch</td><td>點狀按照“項目 II-3 偏光板氣泡”的標準。
Spot type in accordance with the criteria of “Item II-3. Polarizer bubble”.
線狀按照“項目 I-1 玻璃刮傷”的標準。
Line type in accordance with the criteria of “Item I-1. Glass scratch”.</td><td>次要
Minor</td></tr>
<tr><td>2. 表面汙漬
Stains on surface</td><td>表面汙漬無法用軟布或類似的清潔物輕輕擦拭去除。
Stains cannot be removed even when wiped lightly with a soft cloth or similar cleaning.</td><td>次要
Minor</td></tr>
<tr><td>3. 偏光板氣泡
Polarizer bubble</td><td>(mm)
<table>
<tr><th>尺寸
Size</th><th>容許個數
number of pieces permitted</th></tr>
<tr><td>Φ≦0.2</td><td>忽略
Ignore</td></tr>
<tr><td>0.2＜Φ≦0.5</td><td>2</td></tr>
<tr><td>0.5＜Φ</td><td>0</td></tr>
<tr><td>顯示區外
beyond A.A.</td><td>忽略
Ignore</td></tr>
</table></td><td>次要
Minor</td></tr>
</table>

<table>
<tr><th>項目
Item</th><th>描述
Description</th><th>標準
Criterion</th><th>允收
水準
AQL</th></tr>
<tr><td rowspan="5">III. 顯示
Displaying</td><td>1. 耗電
Power consumption</td><td>該模組的工作電流消耗不應超出產品規格書的規範。
The module operating current consumption should not go beyond the standard indicated in Product Specification</td><td>主要
Major</td></tr>
<tr><td>2. 像素尺寸
Pixel size</td><td>顯示像素的尺寸的公差應規格的±25%之內。
The tolerance of display pixel dimension should be within ±25% of specification.</td><td>次要
Minor</td></tr>
<tr><td>3. 顏色
Color</td><td>依據產品規格。
Refer to the product specification.</td><td>主要
Major</td></tr>
<tr><td>4. 亮度
Luminance</td><td>依據產品規格。
Refer to the product specification.</td><td>主要
Major</td></tr>
<tr><td>5. 暗點、亮點 、髒污
Dimming spot、Lighting spot、Dust</td><td>1.
<table>
<tr><th>平均直徑
Average diameter
D:(mm)</th><th>容許個數
number of pieces permitted</th></tr>
<tr><td>D ≦0.1</td><td>忽略
Ignore</td></tr>
<tr><td>0.1 ＜ D ≦0.15</td><td>1</td></tr>
<tr><td>0.15＜ D ≦0.2</td><td>1</td></tr>
<tr><td>0.2 ＜ D</td><td>0</td></tr>
<tr><td>顯示區外
beyond A.A.</td><td>忽略
Ignore</td></tr>
</table>
D=(長邊直徑 + 短邊直徑)/2
D=(long diameter + short diameter)/2
像素暗點是不允許。
Pixel off is not allowed.</td><td>次要
Minor</td></tr>
</table>

<table>
<tr><th>項目
Item</th><th>描述
Description</th><th>標準
Criterion</th><th>允收
水準
AQL</th></tr>
<tr><td>III. 顯示
Displaying</td><td>5. 暗點、亮點 、髒污
Dimming spot 、Lighting spot、Dust</td><td>2.
<table>
<tr><th>寬
width(mm)
W</th><th>長
length(mm)
L</th><th>容許個數
number of pieces permitted</th></tr>
<tr><td>W≦0.03</td><td>忽略
Ignore</td><td>忽略
Ignore</td></tr>
<tr><td>0.03＜ W≦0.05</td><td>L≦1</td><td>3</td></tr>
<tr><td>0.05＜ W</td><td>------</td><td>無
None</td></tr>
<tr><td>顯示區外
beyond A.A.</td><td>------</td><td>忽略
Ignore</td></tr>
</table></td><td>次要
Minor</td></tr>
<tr><td rowspan="3">IV. 軟板
Film</td><td>1. 尺寸
Dimension</td><td>軟板尺寸超規。
Film dimension out of Spec.</td><td>主要
Major</td></tr>
<tr><td>2. 損傷
Damage</td><td>破損；深刮傷；深摺痕；深壓痕或其他損害是不能接受的。
Crack; deep scratch; deep fold; deep pressure mark or other damage is not acceptable.</td><td>次要
Minor</td></tr>
<tr><td>3. 異物
Foreign material</td><td>導電異物附著在導線，軟板和玻璃之間的異物是不能接受的。
Conductive foreign material sticking to the leads, foreign material between film and glass are not acceptable.</td><td>次要
Minor</td></tr>
</table>

14. APPENDIXES

APPENDIX 1: DEFINITIONS

A. DEFINITION OF CHROMATICITY COORDINATE

The chromaticity coordinate is defined as the coordinate value on the CIE 1931 color chart for R, G, B, W.

B. DEFINITION OF CONTRAST RATIO

The contrast ratio is defined as the following formula:

$$\text{Contrast Ratio} = \frac{\text{Luminance of all pixels on measurement}}{\text{Luminance of all pixels off measurement}}$$

C. DEFINITION OF RESPONSE TIME

The definition of turn-on response time Tr is the time interval between a pixel reaching 10% of steady state luminance and 90% of steady state luminance. The definition of turn-off response time Tf is the time interval between a pixel reaching 90% of steady state luminance and 10% of steady state luminance. It is shown in Figure 2.

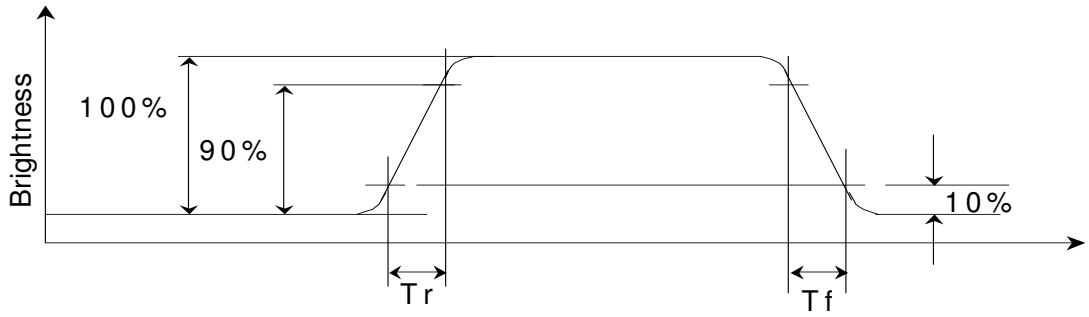

Figure 2 Response time

D. DEFINITION OF VIEWING ANGLE

The viewing angle is defined as Figure 3. Horizontal and vertical (H & V) angles are determined for viewing directions where luminance varies by 50% of the perpendicular value.

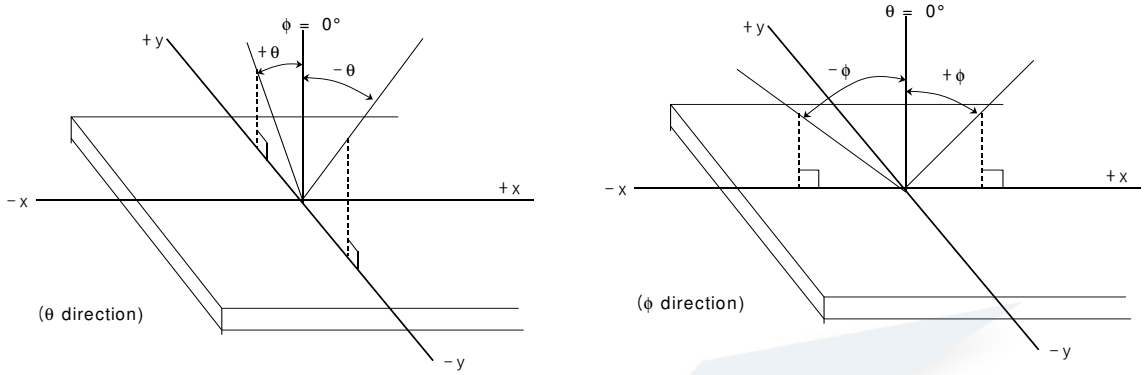

Figure 3 Viewing angle

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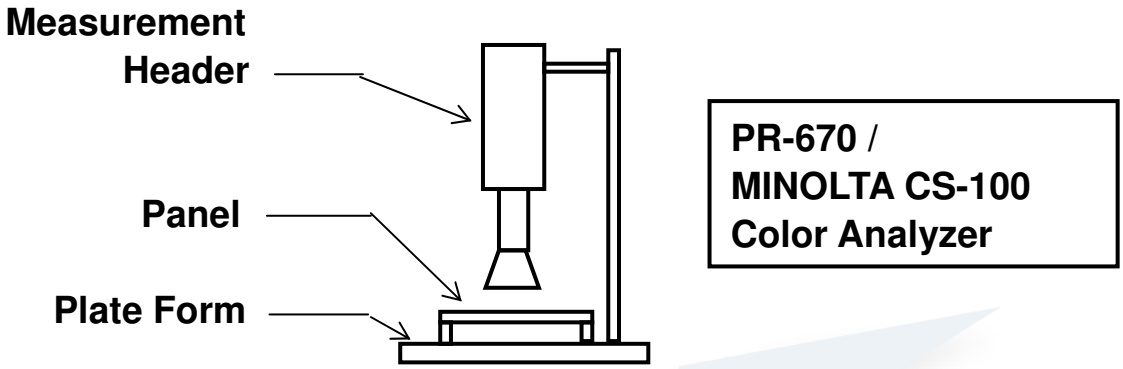
APPENDIX 2: MEASUREMENT APPARATUS

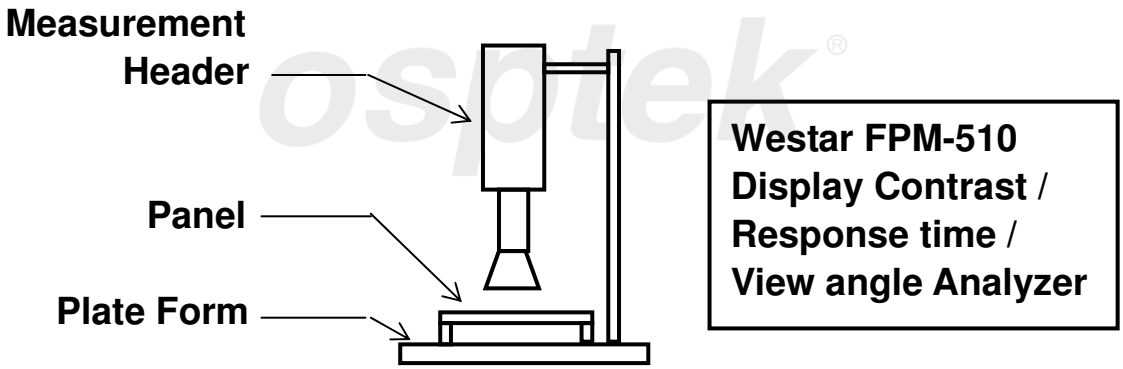
A. LUMINANCE/COLOR COORDINATE

PHOTO RESEARCH PR-670, MINOLTA CS-100

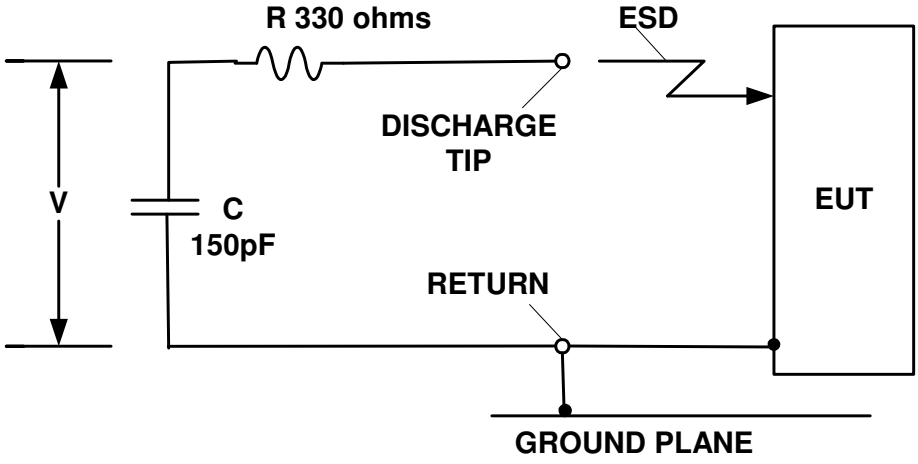


B. CONTRAST / RESPONSE TIME / VIEWING ANGLE

WESTAR CORPORATION FPM-510



C. ESD ON AIR DISCHARGE MODE

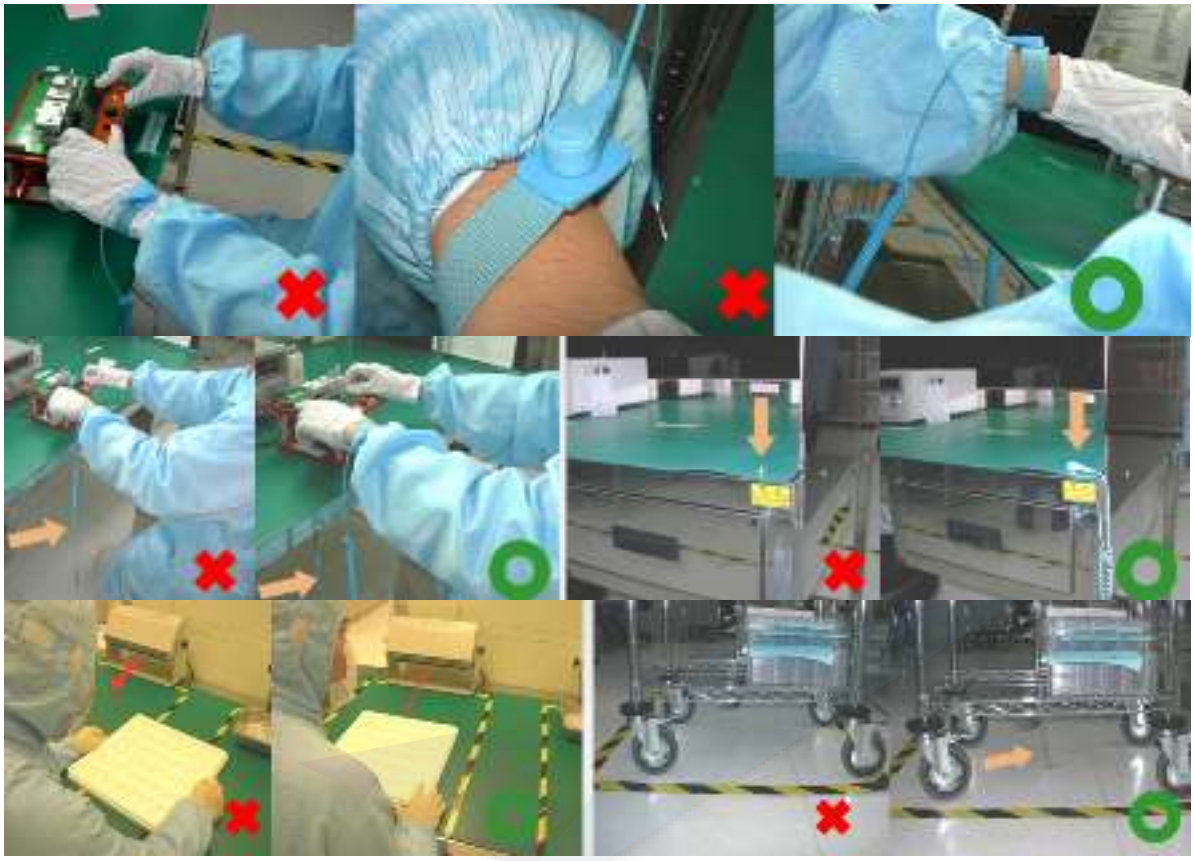


APPENDIX 3: PRECAUTIONS FOR USING THE OLED MODULE

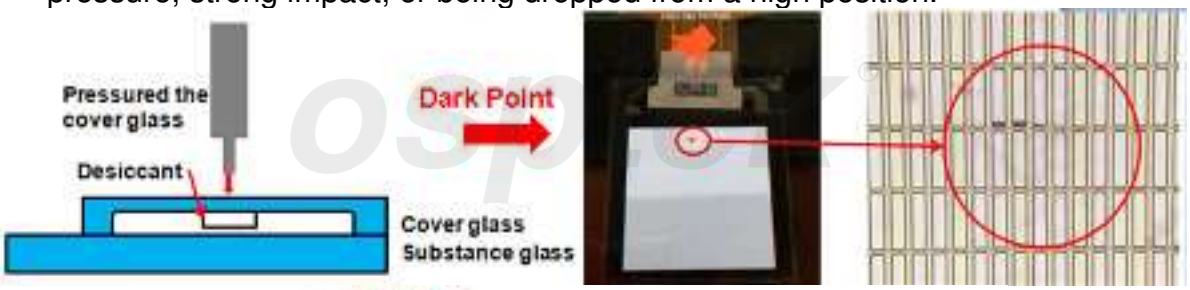
Precautions for Handling

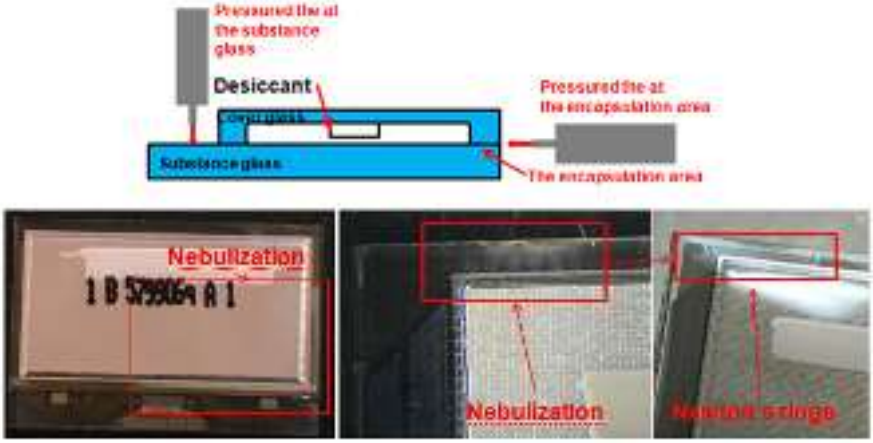
1. When handling the module, wear powder-free anti static rubber finger cots/ anti-static clothing, anti-static gloves ,antistatic wrist strap and anti-static shoes
   The environment should dispose the static elimination blower, anti-static pad, anti-static chair, and anti-static floor. The humidity maintains usually more than 40%

2. The OLED module is an electronic component and is subject to damage caused by Electro Static Discharge (ESD). And hence normal ESD precautions must be taken when handling it. Also, appropriate ESD protective environment must be administered and maintained in the production line. When handling and assembling the panel, wear an antistatic wrist strap with the alligator clip attached to the ground to prevent ESD damage on the panel. Antistatic wrist strap should touch human body directly instead of gloves. (See below photos).



3. The OLED module is consisted of glass and film, and it should avoid pressure, strong impact, or being dropped from a high position.

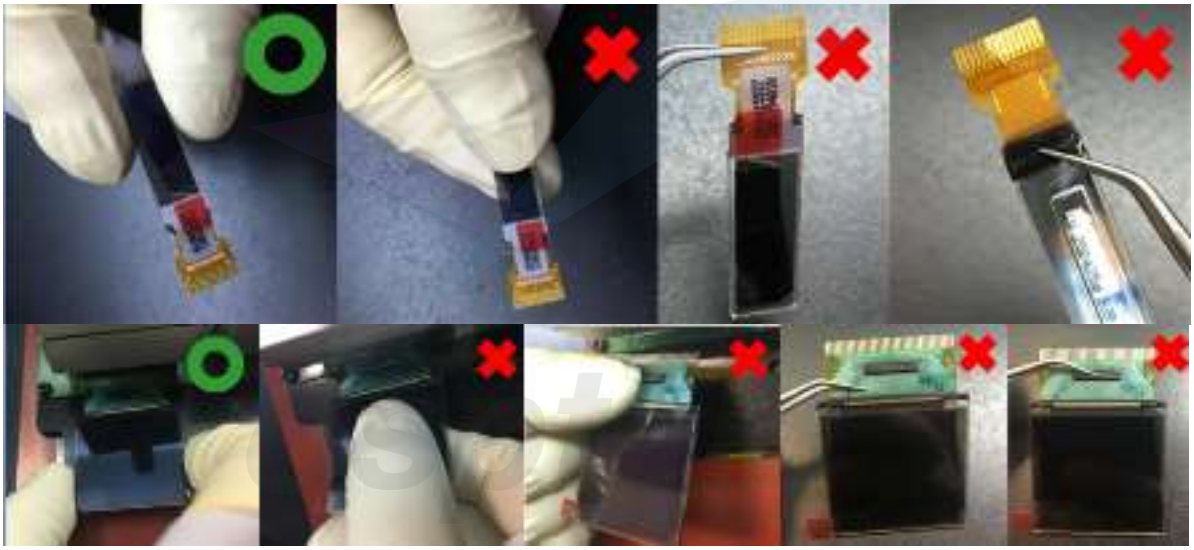




4. Take out the panel one by one from the holding trays for assembly, and never put the panel on top of another one to avoid the scratch.

5. Avoid jerk and excessive bend on TAB/FPC/COF, and be careful not to let foreign matter or bezel damage the film.

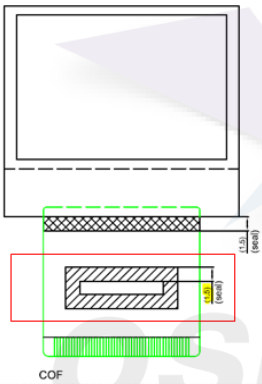
6. When handling and assembling the module (panel + IC), grab the panel, not the TAB/FPC/COF.

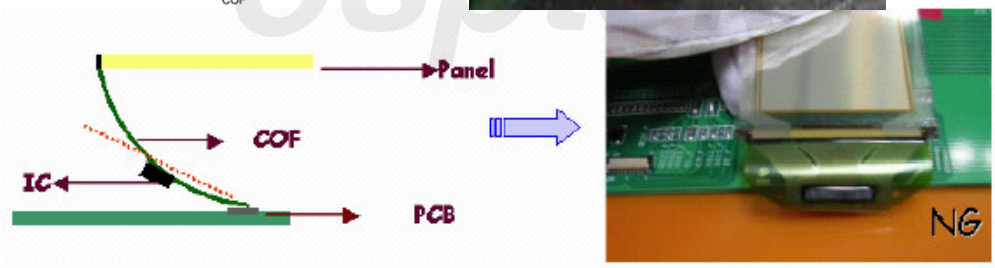


7. Use the tweezers to open the clicks on the connector of PCB before the insertion of FPC/COF, and click them back in. Once the FPC/COF sits properly in the connector, use the tweezers to avoid the damages.

8. Please do not bend the film near the substrate glass. It could cause film peeling and TAB/FPC/COF damage. For TAB, It should bend the slit area as actual OLED it is. For FPC or COF, it is suggested to follow below pictures for instruction (distance between substrate glass and bending area >1.5mm; R>0.5mm).

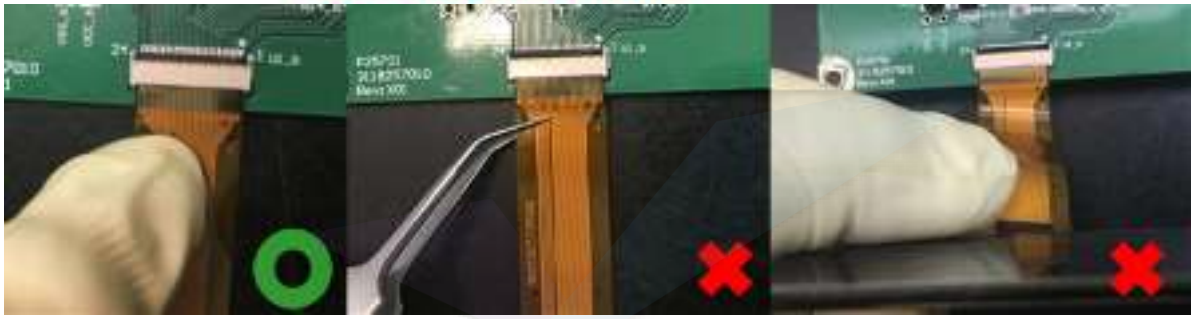
9. Avoid bending the film at IC bonding area. It could damage the IC ILB bonding. It should avoid bending the IC seal area. Please keep the bending distance >1.5mm.





Use finger to insert COF /FPC into the connector when assembling the panel. Please refer to the photo.

COF: Use both thumbs



10. Do not wipe the pin of film and polarizer with the dry or hard materials that will damage the surface. When cleaning the display surface, use the soft cloth with solvent, IPA or alcohol, to clean.

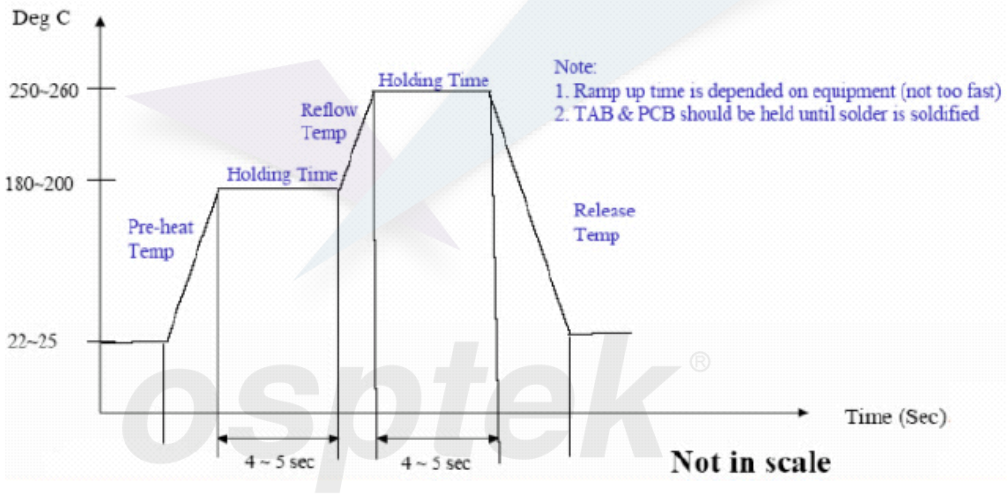
11. Protection film is applied to the surface of OLED panel to avoid the scratch. Please remove the protective film before assembling it. If the OLED panel has been stored for a long time, the residue adhesive material of the protective film may remain on the display surface after remove the protective film. Please use the soft cloth with solvent, IPA or alcohol, to clean.

12. When hand or hot-bar soldering TAB/FPC onto PCB, make sure the temperature and timing profiles to meet the requirements of soldering specification (the specification depends on the application or optimized by customer) to prevent the damage of IC pins by inappropriate soldering, and also avoid the high temperature to damage the Organic light-emitting materials.

13. Solder residues arise from soldering process have to be cleaned up thoroughly before the module assembly.

14. Use the voltage and current settings listed in the specification to do the function test after the module assembly.

15. Suggestion for soldering process:
    i. TAB Lead- free soldering hot bar process
    1. Use pulse heated bonding tool equipment
    2. Material: Sn/Ag/Cu lead-free solder paste with typical 25um thickness on PCB pad. The TAB pin size and shape may be different, please base on the production line to adjust the thickness of PCB pad and temperature.S
    3. Bonding Force:--4kg per centimeter square as the starting point.
    4. Suggested bonding tool temperature & time profile is as below for reference. Since there are differences in TAB soldering pins, soldering technicians' skills, mechanism…etc., the soldering conditions must be adequately tuned.



    ii. TAB Lead- free soldering wire process
    In case of manual soldering (Lead- free solder wire)
    1. Solder wire contact iron directly: 280±5℃ at 3-5secs
    2. Solder wire contact TAB lead directly (near iron but not contact): 380±5 ℃, 3-5secs
    3. Since there are differences in TAB soldering pins, soldering technicians' skills, mechanism…etc., the soldering conditions must be adequately tuned.
    iii. High temperature will result in rapid heat conduction to IC and might cause damage to IC, so please keep the temperature below 380℃. Also, avoid damaging the polyimide and solder resist which might take place at high temperatures. Refold cycles base on the de-soldering status, if the plating of pin was damaged, it can not be used again.

Precautions for Electrical

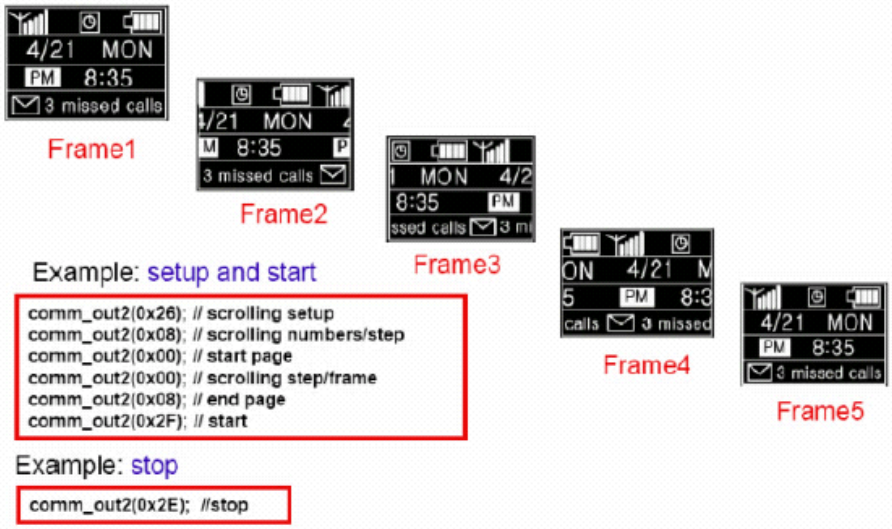
1. Residual Image (Image Sticking)

The OLED is a self-emissive device. As with other self-emissive device or displays consisting of self-emissive pixels, when a static image frozen for a long period of time is changed to another one with all-pixels-on background, residual image or image sticking is noticed by the human eye. Image sticking is due to the luminance difference or contrast between the pixels that were previously turned on and the pixels that are newly turned on. Image sticking depends on the luminance decay curve of the display. The slower the decay, the less prominent the image sticking is. It is strongly recommended that the user employ the following four strategies to minimize image sticking.

1. Employ image scrolling or animation to even out the lit-on time of each and every pixel on the display, also could use sleeping mode for reduced the residual image and extend the power capacity.
2. Minimize the use of all-pixels-on or full white background in their application because when the panel is turned on full white, the image sticking from previously shown patterns is the most revealing. Black background is the best for power savings, greatest visibility, eye appealing, and dazzling displays.
3. Avoid displaying the characters or menu with high brightness level in a fix position for a long time or repeatedly. If necessary, using the auto fadeout technology.
4. If a static logo is used in the reliability test, change the pattern into its inverse (i.e., turn off the while pixels and turn on the previously unlit pixels) and freeze the inverse pattern as long as the original logo is used, so every pixel on the panel can be lit on for about the same time to minimize image sticking, caused by the differential turn-on time between the original and its reverse patterns.

Black Background

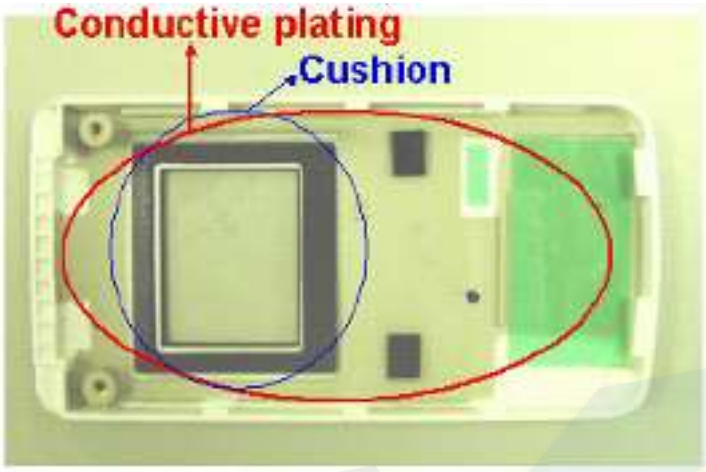
Scrolling example



Precautions for Mechanical

1. Cushion or Buffer tape on the cover glass

It is strongly recommended to have a cushion or buffer tape to apply on the panel backside and front side when assembling OLED panel into module to protect it from damage due to excessive extraneous forces.



It is recommended that a plating conductive layer be used in the housing for EMI/EMC protection. And, the enough space should be reserved for the IC placement if the IC thickness is thicker than the TAB film when customer design the PCB.

2. Avoid excessive bending of film when handling or designing the panel into the product

The bending of TAB/COF/FPC has to follow the precautions indicated in the specification, extra bending or excessive extraneous forces should be avoided to minimize the chances of film damage. If bending the film is necessary, please bend the designated bending area only. Please refer to items 8 and 9 of Precautions for Handling for more information.

Precautions for Storage and Reliability Test

1. Storage

Store the packed cartons or packages at 25℃±5℃, 55%±10%RH. Do not store the OLED module under direct sunlight or UV light. For best panel performance, unpack the cartons and start the production of the panels within six months after the reception of them.

2. Reliability Test

Osptek only guarantees the reliability of the OLEDs under the test conditions and durations listed in the specification.

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